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EXAMEN CODE:

**Specific instructions for the Chemistry Contest**

Highly esteemed student:

- This exam was created by a Scientific Committee composed of former olympic students who work (mostly) at the University of Havana.
- **Complete your personal information.** Please verify with your ID card.
- The direction of the Provincial Education is responsible for ensuring that the exam is printed in a sufficiently high quality, allowing students to read the instructions, questions, figures, data, graphs, and tables. **No questions will be canceled due to poor printing.**
- Write all your information clearly in the space provided (first name, last name, school, municipality, province). **Do not write anything in the CODE field,** it is for the exclusive use of the coding manager.
- **Check that your exam is complete.** In the case of the exam on the second day, 10th grade should answer problems 1, 2, 3 and 4; 11th grade should answer problems 1, 2, 3 and 5; 12th grade should answer problems 1, 2, 5 and 6.
- All necessary constants, conversions, equations, and periodic table are included on the first page of the exam. This page will be removed during the coding process.
- At the beginning of each question, the grading is indicated, its value (out of a total of 100 points), as well as the value of each item (in marks).
- You can use a scientific calculator and a ruler.
- Please comply to the rules of approximate calculation. Attention! You have limited time and the exam has a large number of items. You do not have a space limit to answer, BUT please answer in an orderly manner, with appropriately sized and legible font, to facilitate the grading process.
- You must present all auxiliary calculations, justifying your reasoning and answers properly whenever necessary. **Unintelligible answers will not be graded.**
- Number the additional answer sheets at the bottom of each page and indicate the total number of pages, for example, “1 of 7,” “2 of 7,” and so on.
- If the exam instructor/proctor interferes with any of the above guidelines due to ignorance, exercise your right to protest and immediately request the intervention of the competent authority of the Ministry of Education, which is the institution responsible for administering the exam.
- The President of the Scientific Committee can be reached at +53-5045-4299.

You are a source of pride for Cuban education: Best of luck!

EXAM CODE (only for the use of the code maker)

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal Gas Law	$p \cdot V = n \cdot R \cdot T$
Gas constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero in Celsius scale	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Vapor pression lowering	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Sphere volume	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Sphere area	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Circle area	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Length of a circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Gravity force	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$ $1 \text{ bar} = 100 \text{ kPa}$ $1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$ $1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$ $1 \text{ dm}^3 = 1 \text{ L}$ $1 \text{ mL} = 1 \text{ cm}^3$ $1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

EXAM CODE (only for the use of the code maker)

Problem 7. Carbon and sulfur (15% of total; 10th, 11th and 12th grades)

7.1	7.2	7.3	7.4	7.5	7.6	Total	Total
10	8	4	8	2	28	60	15

Ernesto is a very curious student, and whenever he has the opportunity, he conducts experiments in the school laboratory with technician Ivis and mentor Agustín. After conducting a bibliographic search, the mentor discovered that when carbon and sulfur combine, different binary substances are obtained, and he asked the student to answer the following question.

7.1. What is the formula for the four sulfur compounds of carbon containing approximately 36, 27, and 16% carbon by mass? It is known that in one case, both the monomer and the dimer can exist, and none of the molecules is made up of more than five atoms.

7.2. Help Ernesto to represent the structures of the four molecules described above.

Ernesto liked the challenge and asked permission to try to obtain the substances at the school. But Ivis denied him, arguing that sulfur compounds are toxic and that the reactions involved were highly exothermic. Fortunately, Agustín has excellent relations with the University of Sancti Spiritus, and they planned to perform the reactions under more controlled conditions the following day.

Upon arriving at the university, Ivis found a **10,0 L** rigid reactor equipped with a pressure gauge and placed it under a fume hood. Ernesto weighed 60,0 g of a carbon-sulfur mixture on a technical balance and placed it inside the container. At that moment, Agustín sent them out for a snack, and when he was left alone in the laboratory, he secretly modified the experimental conditions. Upon returning, the student pressed a button, producing a spark and a small explosion inside the reactor. After cooling to room temperature, Ernesto was shocked and disappointed because the pressure inside the container was back to where it had been. Upon careful opening, he noticed a very unpleasant odor and the absence of solid residue. Then the mentor confessed: he had removed most of the solid mixture and filled the container with oxygen.

7.3. Can you help Ernesto to formulate the chemical reactions that happened?

“Agustín, you’re always so reckless!” Ivis said angrily and expelled him from the room. Ernesto weighed the portion of the carbon-sulfur mixture his mentor had hidden. He found a mass of only **57,3 g** and poured it completely out of the reactor. The technician made sure to evacuate all the air and replace it this time with the inert gas argon, until a pressure of 1,00 atm was reached. A spark was then ignited, and a strongly exothermic reaction occurred inside the vessel, resulting in the formation of substances **A** and **B** in a gaseous state. No solid residue was observed through the glass window of the apparatus. The system was then placed in an ice bath. After cooling to 100 °C, the pressure inside the vessel was 3,81 atm. Cooling continued, and upon reaching the temperature of 25 °C, a few drops appeared on the window due to the condensation of some of product **B**, and the pressure dropped to 1,96 atm. Upon reaching 0 °C and after carefully evacuating the gases (only Ar and product **A**), between 37 and 38 mL of a clear, volatile liquid with a very unpleasant odor were recovered (all of product **B**), which, when measured with a hydrometer, had a density of 1,29 g/mL.

7.4. Show that the amount of oxygen initially introduced into the reactor by the mentor was sufficient to react with the small portion of the mixture that was combusted.

Vapor pressure is the pressure exerted by the portion of a volatile substance in the gaseous phase, in direct contact –that is, in equilibrium– with the corresponding volatile liquid or solid. Vapor is said to be “saturated” when the gaseous phase contains the largest amount of the volatile substance thermodynamically possible.

7.5. The saturated vapor pressure of **B** at 25 °C is 48,1 kPa. Calculate the maximum amount of **B** in the gas phase that the reactor can hold at that temperature.

7.6. Help the student to calculate the identity of **A** and **B**.

Note: The experiments described in this problem are entirely fictitious and extremely dangerous. **Never** attempt to reproduce them.

Problem 8. Benzoyl peroxide (10% of total; only 10th and 11th grades)

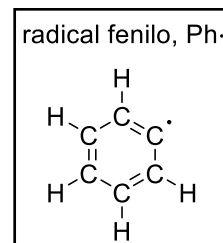
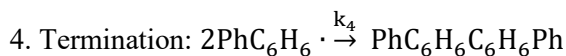
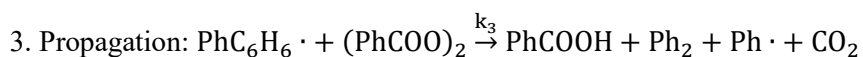
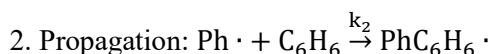
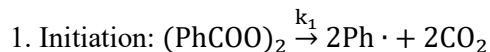
8.1	8.2	8.3	8.4	8.5	8.6	8.7	Total	Total
2	10	4	10	2	4	8	40	10

Organic peroxides decompose easily, generating free radicals, which is why they are used as initiators in chain reaction, especially in industrial polymerization processes. One day, Bárbaro and Pedro studied the decomposition of benzoyl peroxide $(\text{PhCOO})_2$ by refluxing in the solvent benzene. These students work well as a team, as their skills complement each other: the former has solid theoretical knowledge, and the latter is very capable in the laboratory.

Pedro's measurements demonstrated that the rate law is first-order with respect to peroxide. However, the values of the reaction rate constants k_{obs} were found to be dependent on the initial concentration of benzoyl peroxide $[\text{PB}]_0$, as shown in the data in the following table.

$[\text{PB}]_0$ (mol/L)	0,0205	0,0830	0,156	0,225	0,283
k_{obs} (10^{-5} s^{-1})	3,85	5,30	6,35	7,10	7,70

These results suggested that, in addition to the unimolecular thermal decomposition of benzoyl peroxide, decomposition processes mediated by some free radical intermediate also occur. Bárbaro found that the following simplified decomposition mechanism is reported in the literature.



8.1. Why is the O-O single bond present in peroxides relatively weak?

Bárbaro applied the Steady State Theory to the mechanism and showed that the rates of the initiation step 1 and termination step 4 of the mechanism have to be equal, and that the rate of decomposition of the peroxide obeys a complex expression of the form $v_r = -\frac{d[\text{PB}]}{dt} = (\mathbf{A} + \mathbf{B} \cdot \sqrt{[\text{PB}]}) \cdot [\text{PB}]$, where the constants **a** and **b** can be expressed in terms of the specific rate constants of each step.

8.2. Deduce the complete reasoning that Barbaro followed.

8.3. The values of the specific rate constants are $k_1 = 2,4 \cdot 10^{-5} \text{ s}^{-1}$ and $k_4 = 2,0 \cdot 10^{11} \text{ u}$. State the units of measurement “u” of k_4 and calculate the concentration of the radical $\text{PhC}_6\text{H}_6 \cdot$ if we start from an initial peroxide concentration of 0,010 mol/L.

8.4. Fit Pedro's results in the table to a straight line and calculate the values of the constants **A** and **B** of Barbaro's rate equation

$[\text{PB}]_0$ (mol/L)	0,0205	0,0830	0,156	0,225	0,283
k_{obs} (10^{-5} s^{-1})	3,85	5,30	6,35	7,10	7,70

8.5. When Pedro asked Bárbaro why the fourth reaction step (termination) is the fastest of all, what do you think his answer was?

8.6. At low concentrations of peroxide $k_{\text{obs}} = k_1$, while at high concentrations of peroxide $k_{\text{obs}} = k_3 \sqrt{\frac{k_1}{k_4}}$. Use the

Arrhenius model and express the experimentally observed activation energy for the reaction E_A^{obs} as a function of the activation energies of the individual steps for both cases.

8.7. Estimate the minimum values of the activation energies E_{A1} , E_{A2} , E_{A3} and E_{A4} .

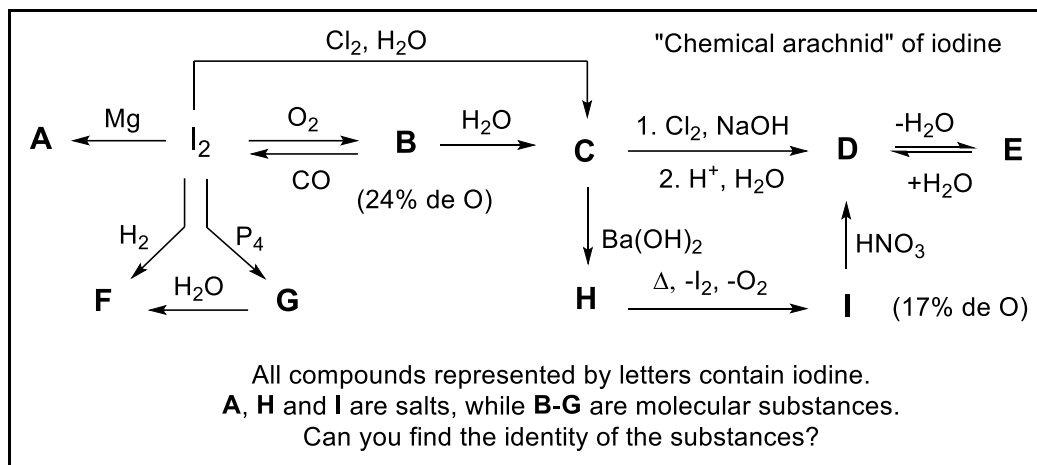
enlace	H-H	C-C	O-O	C-H	C-O	O-H	C=C	C=O
E (kJ/mol)	436	348	146	413	358	463	612	799

Problem 9. Iodine (15% of total; only 10th and 11th grades)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9	Total	Total
20	11	7	2	2	5	4	4	5	60	15

When Melissa discovered that Cynthia was afraid of spiders, she came up with a plan to help her overcome her fears. She took a piece of paper that was black on one side and white on the other, wrote a note on the white side, and folded it to create an origami arachnid. Melissa then pretended to be holding a real spider, and Cynthia began to scream in terror –until Melissa confessed–, “Hold it. It won’t

bite you; it’s a chemical spider.” After gathering a lot of courage, Cynthia took the origami paper and found the following problem written on the white side of the paper sheet.



9.1. Help Cynthia to find the formulas for substances A-I. Record your calculations.

Hint: D and E are iodine oxoacids in the same oxidation state. The molecule of D has a square-base bipyramid geometry, in which five I-O bonds measure 1,87-1,91 Å, and one I-O bond measures 1,78 Å.

9.2. Draw the structures of the iodine species shown in Melissa’s note. Note the geometry according to the postulates of Gillespie’s theory (TRPEV).

9.3. Melissa discovered in an article that the reaction between diiodine and A in concentrated sulfuric acid is complex and results in the formation of the species IO⁺, I₂O, I₃⁺, I₅⁺. Draw their structures.

9.4. At room temperature, iodine is solid, while bromine is liquid, but chlorine and fluorine are gases. Why?

9.5. Hydrogen iodide behaves like a strong acid when dissolved in water, completely dissociating into ions. In contrast, hydrogen fluoride behaves like a weak acid and only partially dissociates. Why is this happening?

9.6. Describe the diiodine molecule using Molecular Orbital Theory and state: i) electron distribution, ii) bond order, iii) magnetic properties, and iv) frontier orbitals.

9.7. Iodine reacts with the sulfite anion to form sulfate and iodide ions. Propose a probable mechanism for the reaction, indicating the movement of electrons with arrows.

On the day of the science fair, Cynthia decided it was time for revenge. While Melissa was studying unsuspectingly, Cynthia threw some ammonia and iodine in the hallway, right outside the lab entrance, and waited patiently. When the break time bell rang, explosions and screams were heard. It was Melissa, scared, not understanding why the floor was exploding beneath her shoes. And the rest of the chemistry group was laughing out loud, well aware of their mentors’ favorite joke. The explosive compound that forms *in situ* during a chemical reaction has the formula NI₃·NH₃.

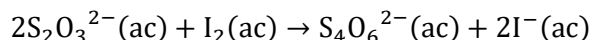
9.8. Propose a probable mechanism for the reaction of nitrogen triiodide formation from ammonia and diiodine, indicating the movement of electrons with arrows.

9.9. Draw the structure of nitrogen triiodide ammonate. Explain why this compound is extremely unstable and detonates at the slightest vibration or contact. Write the equation for the decomposition reaction that occurs during the explosion.

Problem 11. Iodometry or iodimetry (20% of total; only 11th and 12th grades)

11.1	11.2	11.3	11.4	11.5	11.6	11.7	11.8	11.9	11.10	11.11	11.12	11.13	11.14	11.15	11.16	Total	Total
2	4	2	2	6	2	7	2	8	4	8	4	9	4	9	7	80	20

Mentor Orestes (better known by his surname Landrove) knows that iodine methods are suitable for performing quantitative determinations of certain analytes. Although it is possible to perform the titration directly using iodine, he always recommends using an alternative titrant, typically sodium thiosulfate, which is a mild reducer capable of reacting with the iodine formed by an analyte. The main reaction that occurs is



11.1. One of the reasons why iodine is not preferred as a titrant is that it is very volatile. To counteract this difficulty, it is common to add excess iodide ions. Why?

11.2. Calculate the equilibrium constant of the main reaction of the titration at 25 °C.

11.3. The reaction above is based on the high affinity of the sulfur in thiosulfate for iodine. Explain this phenomenon according to the Hard and Soft Acids and Bases Theory.

11.4. The stability of thiosulfate solutions is very sensitive to the pH of the medium. For example, acidic impurities cause cloudiness and a pungent odor. Write the equation for the reaction that occurs.

During the training prior to the International Chemistry Olympiad, Landrove prepared several experiments based on diiodine titrations for Hector. First, they attempted to determine the vitamin C (ascorbic acid, $\text{C}_6\text{H}_8\text{O}_6$, abbreviated as AA hereafter) content in the juice of various fruits. Since ascorbic acid (AA) is not generated through any human metabolic pathway, consuming foods rich in vitamin C, such as citrus fruits, tomatoes, and potatoes, is necessary to prevent diseases like scurvy, which is characterized by collagen deficiency. Hector decided to quantify AA using a redox titration with a standard iodine solution, forming iodide and dehydroascorbic acid ($\text{C}_6\text{H}_6\text{O}_6$). Starch is used as a visual indicator, which forms a deep blue compound in the presence of traces of iodine.

11.5. What masses of iodine and potassium iodide does Hector need to weigh to prepare one liter of aqueous solution with molar concentrations of 0,00100 M and 0,0500 M, respectively?

11.6. Formulate the equation for the reaction between iodine and ascorbic acid.

11.7. In general, the pH of the I_2 solution is adjusted to avoid disproportionation of this species. Which medium is preferred? Justify numerically.

Landrove macerated a variety of fruits and vegetables separately in distilled water using a mortar and pestle. He then filtered the mixture, washed the solid portion, and transferred the collected liquid to a 50-mL flask, topping up with additional filtrated liquid to the required volume. Hector then transferred 10,0 mL of the solution to an Erlenmeyer flask using a volumetric pipette. He added approximately 20 mL of distilled water and 2 mL of 1% starch and titrated with the 0,00100 M iodine standard solution until the first sign of a permanent blue color appeared. The results are in the table.

Food	orange	plump	pepper	guava	parsley
V(I_2), mL	12,5	9,8	16,2	20,0	8,6

11.8. What glassware is needed to perform a gravity filtration?

11.9. Typically, AA content ranges from 10-30 mg per 100 mL of orange juice. Do Hector's results correspond to the values reported in the literature?

Landrove then explained to Hector that the concentration of the standard solution he had prepared was probably incorrect and suggested using back titration to determine the ascorbic acid content. These methods generate an excess of iodine by reacting with an oxidizing primary standard. Some of the iodine formed reacts with the analyte, and the excess is titrated with thiosulfate.

11.10. Formulate the reactions corresponding to the reactions of iodide ions with dichromate and iodate ions in an acidic medium.

11.11. The guava juice that Hector analyzed contained 35,2 mg of AA per 100 mL. Considering that only AA contributes to the acidity of the food, estimate the pH of the solution.

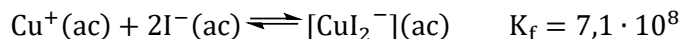
At the end of the session, the mentor tested the student's knowledge of chemical equilibrium. Landrove explained that the copper(II) ion is not capable of oxidizing iodide ions, as evidenced by the standard electrode potential values. However, the mentor dissolved some copper sulfate pentahydrate crystals in pure water in a test tube. Upon adding a portion of potassium iodide and stirring, the solution immediately changed color, first to green and then to dark yellow, and a precipitate appeared. After filtering and washing with water, the precipitate turned white.

11.12. Help Hector explain why the redox reaction of copper(I) iodide precipitate formation occurs, taking into account the polarization of ions and the Hard and Soft Acids and Bases Theory.

11.13. Hector found it easy to demonstrate numerically that in the presence of excess iodide ions, the process described above is enhanced. Can you do it?

11.14. Calculate the solubility (in mol/L) of copper(I) iodide in water.

11.15. Calculate the solubility (mol/L) of copper(I) iodide in a 0,50 mol/L aqueous solution of potassium iodide. Consider the following additional equilibrium that is established



11.16. Explain why copper(I) ions are unstable in aqueous solution using electrode potential values (quantitative) and the Theory of Hard and Soft Acids and Bases (qualitative).

The dissociation constants of ascorbic acid are $\text{pK}_{a1} = 4,1$ y $\text{pK}_{a2} = 11,8$ $\text{pK}_{ps}(\text{CuI}) = 11,9$
 $E^\circ(\text{V}): \text{S}_4\text{O}_6^{2-}/\text{S}_2\text{O}_3^{2-} = 0,08; \text{I}_2/\text{I}^- = 0,54; \text{IO}^-/\text{I}^- = 0,49; \text{HIO}/\text{I}_2 = 1,45; \text{Cu}^{2+}/\text{Cu}^+ = 0,15; \text{Cu}^{2+}/\text{Cu} = 0,34$

ANSWER FROM HERE ON
